AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2013

Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a. Find the highest common factor (H.C.F) of $\left(x^4 - y^4\right)$, $\left(x^2 - y^2\right)$, $(x+y)^2$ and $x^2 + xy$.

b. Simplify $\frac{x-3}{3x-3} - \frac{2(x+1)}{x-1}$.

Q.2. (Total 3 Marks)

If $\frac{3x-4}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$, then find the value of *B*.

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

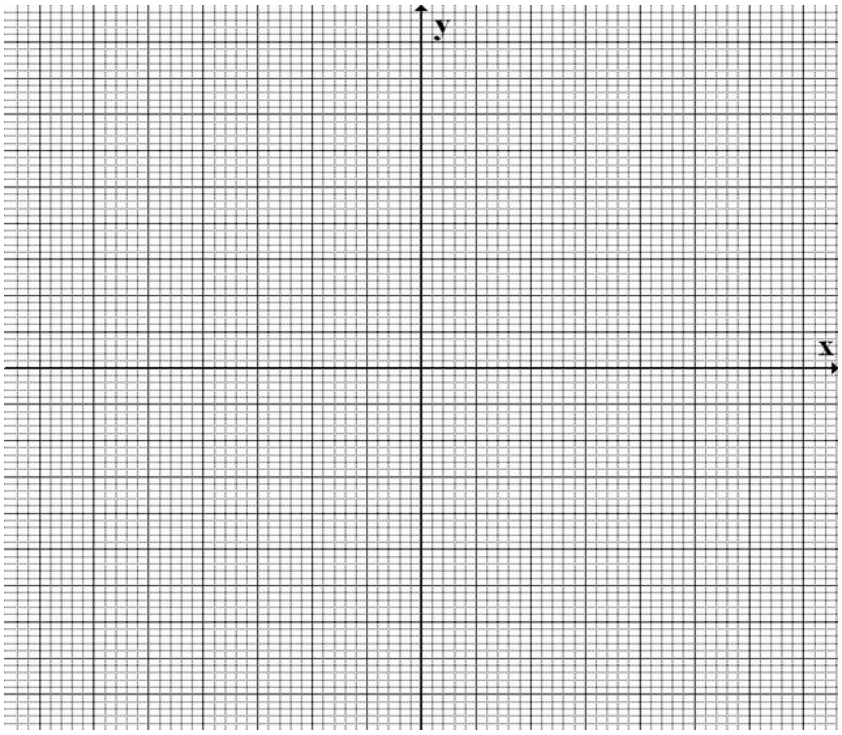
Q.3. (Total 5 Marks)

a. Solve $x - \frac{1}{5} = 3\left(x - \frac{5}{3}\right) - 5x$.

b. Solve $|4x - 1| = |3x - 2|$. Also verify your answer.

Q.4. (Total 5 Marks)

Locate and label $(4,4)$, $(-4,4)$, $(4,-4)$ and $(-4,-4)$ on the given graph. Also find the length of the sides of the geometrical shape obtained.



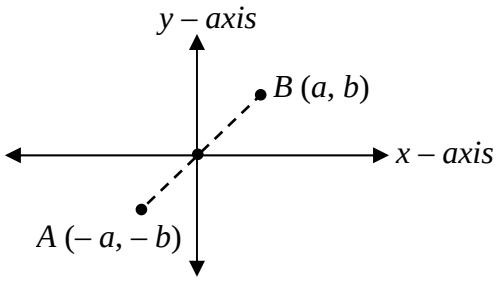
Q.5. (Total 4 Marks)

Solve $3x^2 - 4x + 2 = 0$ by using the quadratic formula.

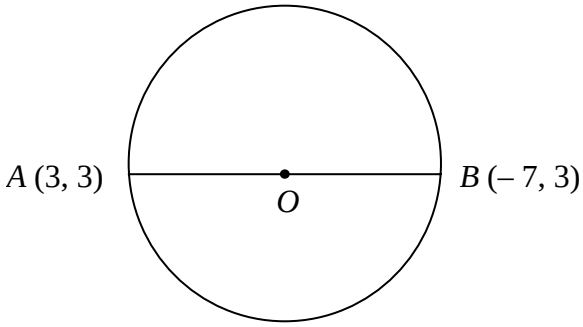
(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 3 Marks)

a. Show that $A\ (-a\ ,-b)$ and $B\ (a, b)$ are equidistant from origin $O\ (0, 0)$.



b. If AB is the diameter of the given circle, then find the coordinates of its centre. Hence, state the quadrant in which the centre of the circle O lies.

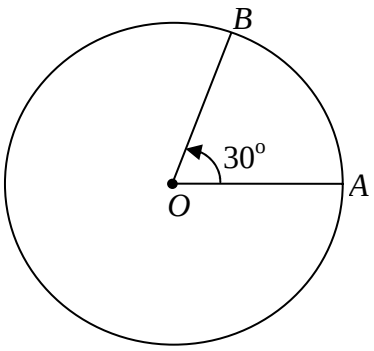


NOT TO SCALE

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a. Consider the given circle with centre *O* and radius 3 cm. Find the length of the arc *AB*.



NOT TO SCALE

b. Correctly match the items in column A with items in column B by an arrow $(\rightarrow)$.

Column A	Column B
$\sec\theta$	$\sqrt{3}$
$\sin 210°$	$\sqrt{1+\tan^2\theta}$
$\tan 60°$	– ve
$\cos 30°$	$\sin 60°$

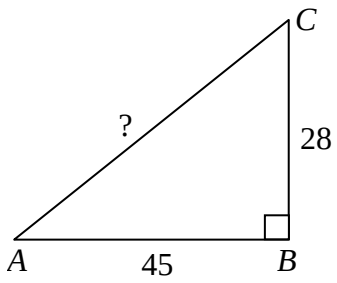
(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 4 Marks)

a.

i. Given that the sides of a triangle measure 24, 8 and 25. Is it a right-angled triangle? Justify your answer. (2 Marks)

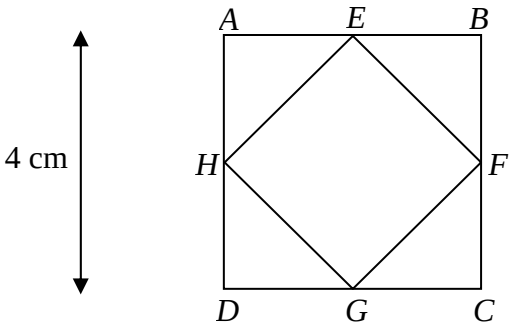
ii. Find the unknown side of the given triangle. (2 Marks)



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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

b. In the given figure *ABCD* is a square and *E*, *F*, *G* and *H* are the mid-points of its sides.

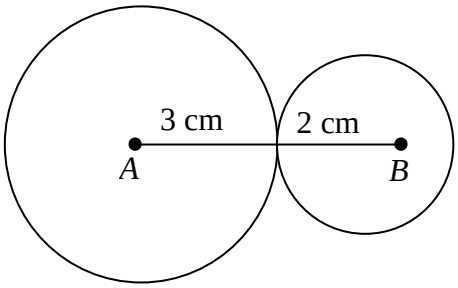


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Are the sides of the square *EFGH* half in measure of the sides of square *ABCD*? Justify your answer by using Pythagoras theorem. Also find the length of diagonals of *EFGH*. (4 Marks)

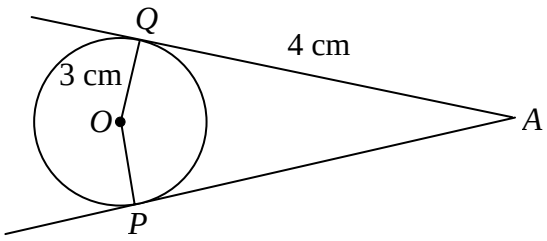
Q.9. (Total 4 Marks)

i. For the given circles having centre at *A* and *B*, find the distance between their centres. (1 Mark)



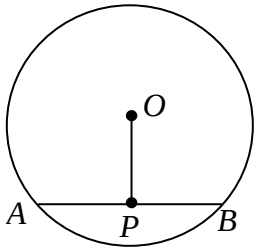
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ii. In the given diagram, find $m\angle OPA$. (1 Mark)



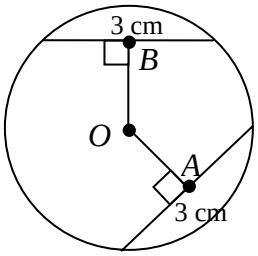
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iii. In the given diagram if $m\ \overline{AB} = 8$ cm, then find $m\ \overline{AP}$. (1 Mark)



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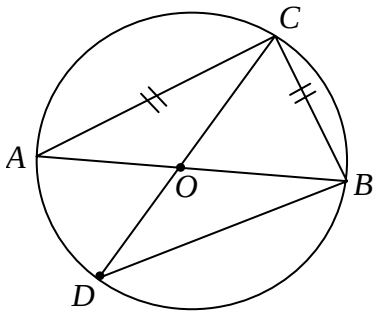
iv. In the given diagram if $m\ \overline{OB} = 2$ cm, then find $m\ \overline{OA}$. (1 Mark)



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Q.10. (Total 4 Marks)

In the given figure *O* is the centre and *AB* is the diameter of the circle. Find $m\angle CDB$ and give a reason to justify each of your steps.



Q.11. (Total 4 Marks)

Draw an equilateral triangle *ABC* with each side measuring 3 cm. Also draw a circumscribed circle of the triangle.

END OF PAPER

Please use this page for rough work